

	Risk-Averse	Self-Adaptive	Risk-Seeking
TSLA		✓	
NFLX		✓	
AMZN		✓	
MSFT			✓
COIN		✓	

Figure 8: The optimal risk inclination for FINMEM when trading different stocks.

To further assess FINMEM’s adaptability to limited training data, we narrowed the training period down to an even shorter period, spanning from August 17, 2021, to February 10, 2022. We then extended the testing phase to cover from February 11, 2022, to April 25, 2023. This evaluation focused on the stock of Tesla, Inc., which has the largest volume of news data. The trading performance of FINMEM during this period is depicted in Figure 7. Remarkably, using less than six months of daily frequency data for training, which encompassed the publication of one Form 10-K and one Form 10-Q, FINMEM consistently ranked high in gains and attained the highest cumulative return after the latter half of December 2022.

The consistently strong trading performance of FINMEM can be attributed to its innovative profiling and memory module design. This design enables FINMEM to effectively integrate, comprehend, and prioritize key information from both textual and numerical data. The flexibility of FINMEM’s profiling module in selecting risk inclinations plays a pivotal role in its ability to both exploit rising market trends and safeguard assets during downturns. A prime example is TSLA, which achieved its best trading results under a self-adaptive risk setting in FINMEM. This configuration enables FINMEM to pursue a conservative and cautious strategy when facing negative short-term cumulative returns. On the other hand, with positive short-term returns, FINMEM switches to an optimistic and assertive approach, thus avoiding excessive passivity. This self-adaptive risk inclination proved effective for most stocks, apart from MSFT as shown in Table 8. For MSFT, a risk-seeking inclination was most beneficial, resonating with its general bullish in the stock market. Additionally, the memory module’s core features, including varied retention times for different information types and critical memory transitions, equip FINMEM to capture essential information for well-informed investment decisions.

6 Ablation Studies

We conducted three distinct ablation studies to evaluate key component alternatives in FINMEM. These studies concentrated on the backbone algorithm, the memory module’s cognitive capacity, and the character setting in the profiling module, particularly examining the aspect of risk inclination. These studies were done using the stock of Tesla, Inc., with a more compact training period from March 14, 2022, to June 15, 2022, and a testing period from June 16, 2022, to December 28, 2022. This shorter duration was chosen for budgetary efficiency, yet it remains sufficient to differentiate the functionality of each component.

6.1 FINMEM Backbone Algorithm Comparison (RQ3)

Metric	B&H	GPT 3.5-Turbo	GPT4	GPT4-Turbo	davinci-003	Llama2-70b-chat
Cumulative Return (%)	-66.9497	16.1501	62.6180	54.6958	1.6308	-52.7233
Sharpe Ratio	-2.0845	2.1589	2.2251	2.4960	0.8515	-2.8532
Daily Volatility (%)	3.8050	0.8862	3.3339	2.5960	0.2269	2.1891
Annualized Volatility (%)	60.4020	14.0683	52.9237	41.2100	3.6018	34.7503
Max Drawdown (%)	67.3269	1.1073	17.4012	12.5734	0.8408	44.7168

Table 3: Comparison of trading performance during the testing period for FINMEM using different LLMs as backbone algorithms.

In our first study, we evaluated the trading performance of FINMEM using various LLMs as its backbone algorithms. The LLMs under consideration included davinci-003, GPT 3.5-Turbo, GPT4, GPT4-Turbo, and Llama2-70b-chat. The

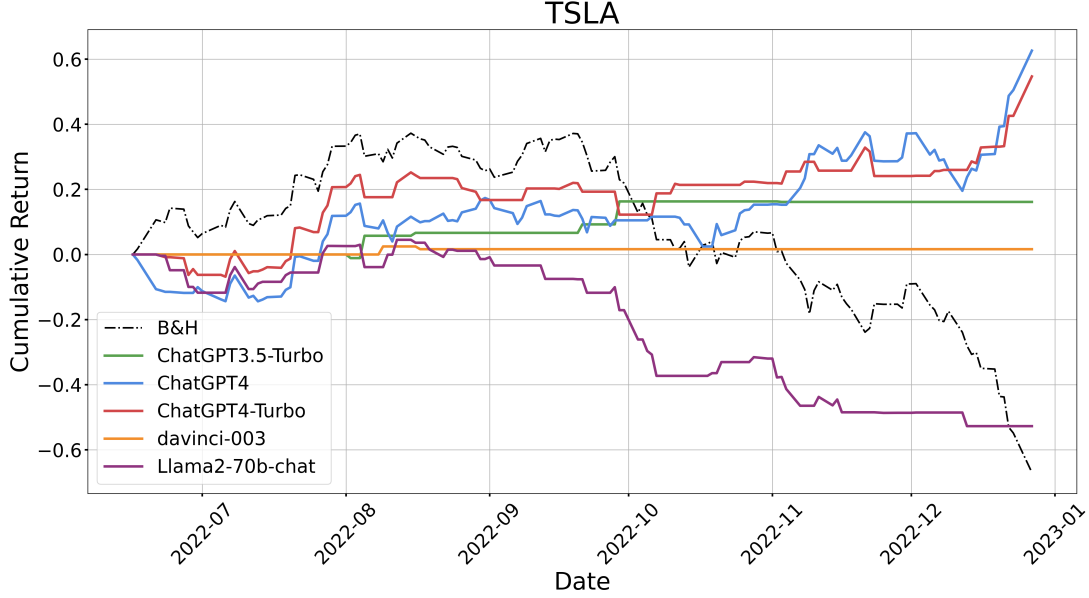


Figure 9: Comparison of overall Cumulative Returns over time for FINMEM using different LLMs as backbone algorithms.

parameter settings were consistent with its optimal performance in the comparative experiment detailed in Section 5, and the risk inclination was configured to be self-adaptive. All other model settings were maintained as outlined in Section 5.1. The results of this evaluation are compiled in Table 3.

In response to **RQ3**, Addressing Research Question 3, the findings demonstrate that FINMEM, powered by GPT-4 and GPT-4 Turbo, delivered superior trading results during the test phase. Specifically, GPT-4 recorded the highest cumulative return, while GPT-4 Turbo exhibited the most favorable Sharpe Ratio. GPT 3.5-Turbo’s performance was also noteworthy, following closely behind. As depicted in Figure 9, though slightly lower than market baseline (B&H), FINMEM with GPT-4 Turbo led in cumulative returns before October 2022. This period was characterized by relative stability and a modest upward trend in TSLA stock. After October 2022, with TSLA undergoing increased volatility and a notable downward trend, the cumulative return trajectory for FINMEM with GPT-4 Turbo exhibited significantly lower volatility and sustained stable returns not markedly lower than those of GPT-4. These results indicate that GPT-4 Turbo is the most suitable backbone algorithm for FINMEM.

FINMEM configured with davinci-003 and Llama2-70b-chat exhibited the lowest Annualized Volatility and Max Drawdown, yet their Cumulative Return and Sharpe Ratio were underwhelming. As illustrated in Figure 9, both models defaulted to a “Hold” strategy beyond a certain point during periods of intense fluctuation in TSLA stock. The unsatisfactory performance of davinci-003 may be attributed to its limited capability, as an earlier generation language model, to capture and understand nuanced yet decisive information.

We selected Llama2-70b-chat as it was deemed to possess stronger in-context learning and instruction-following capabilities compared to other Llama family models with fewer parameters, as noted in Zhao et al. [58]. Nonetheless, in the context of stock trading, it still demonstrated challenges in adequately comprehending key messages necessary for effective trading decisions. The comparatively poorer performance of Llama2-70b-chat can also be attributed to its shorter context window, especially when compared to the GPT models. When integrated with FINMEM, it needs to simplify prompts and shorten the length of retrieved memory insights, which could potentially result in some loss of context. The exceptional trading result demonstrated by GPT-4 Turbo across all models was a main factor in choosing it as the backbone algorithm for FINMEM in our earlier comparative analysis with other algorithmic trading agents.

6.2 Influence about varying the FINMEM character design (RQ4)

In our second study, we focused on evaluating the influence of FINMEM’s profiling module on its trading effectiveness. Specifically, our assessment centered on the effects of customizing trader profiles according to specific stock trading, with a particular focus on risk inclination. As depicted in Figure 8, we equipped FINMEM with three distinct risk

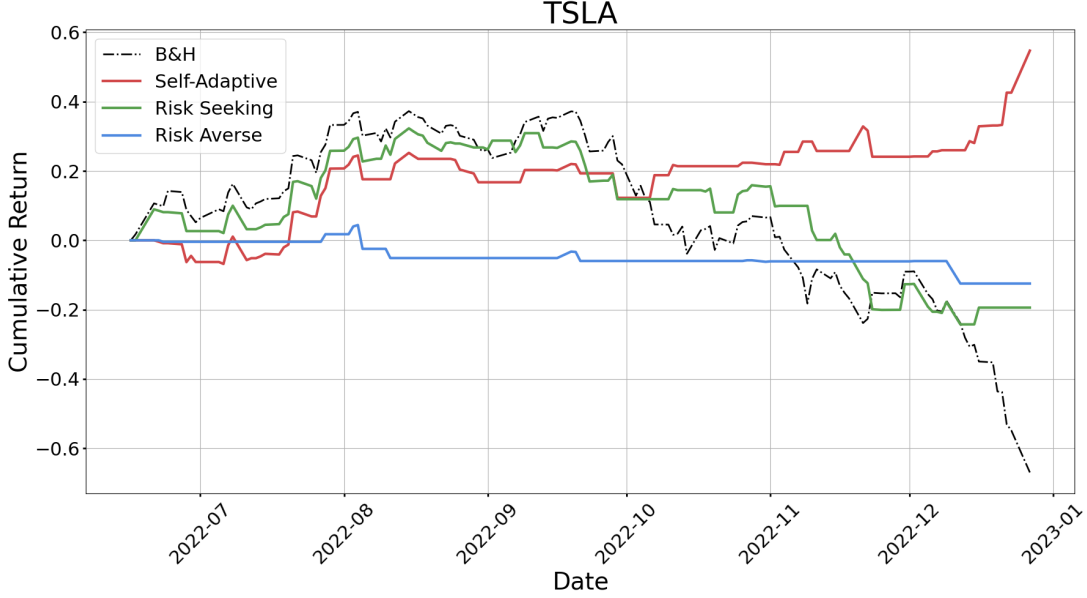


Figure 10: Comparison of Cumulative Return during with different risk inclinations setting in FINMEM’s profiling module.

profiles: risk-seeking, risk-averse, and a self-adaptive character. We executed a comparative analysis of FINMEM’s performance across these risk profiles, maintaining consistency in all other settings as outlined in Section 5.1.

In response to **RQ4**, Table 4 delineates the varied trading performance across different risk profiles. The self-adaptive profile enabled FINMEM to achieve the most favorable trading performance, as it was the only one to secure a positive Cumulative Return and a Sharpe Ratio exceeding 2.0, along with the least Max Drawdown by the end of the testing phase. Figure 10 illustrates FINMEM’s capacity to adeptly navigate substantial stock price volatility and to strategically modulate its trading behavior when necessary. In contrast, the risk-seeking profile, while beneficial during a stable or bullish market as evidenced by MSFT’s performance in Figure 5, exhibited increased volatility and a decline in the face of a market downturn. The risk-averse profile, on the other hand, maintained a more conservative stance, often opting to hold positions. This approach resulted in a Cumulative Return trajectory that generally lagged behind the market baseline, reflecting a degree of overcaution that limited trading activity and potential gains, particularly in a bullish market.

Metric	B&H	Self Adaptive	Risk Seeking	Risk Averse
Cumulative Return (%)	-66.9497	54.6958	-19.4132	-12.4679
Sharpe Ratio	-2.0845	2.4960	-0.7866	-1.5783
Daily Volatility (%)	3.9527	2.7419	3.2722	1.7744
Annualized Volatility (%)	3.8050	2.5960	2.9236	0.9358
Max-Drawdown (%)	67.3269	12.5734	45.0001	15.9882

Table 4: Comparison of overall trading performance during the testing period with different risk inclinations setting in FINMEM’s profiling module.

6.3 Impact of adjusting the capacity of FINMEM working memory (RQ5)

In our third study, we explored whether appropriately tuning the memory retrieval bandwidth of FINMEM could enhance its trading performance. This bandwidth is tied to the working memory’s capacity within its memory module. As depicted in Figure 2, FINMEM retrieves the top- K memory events from its long-term memory in response to a trading inquiry. The working memory capacity is thus set at $3 \times K$, mirroring the human cognitive system’s limit of processing immediate messages upon specific stimuli (3 refers to the three processing layers in long-term memory). By varying the